

A NEAR-LINEAR-TIME SOLVER FOR BILEVEL CONGESTION NETWORK DESIGN VIA DIRECTED NONLINEAR LAPLACIAN FLOW*

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Abstract. Bilevel traffic network design—optimizing link tolls or capacities over a lower-level user equilibrium—is the workhorse formulation of congestion pricing, capacity planning, and origin–destination calibration. It is nonconvex and *NP*-hard, and every practical method solves it by an outer optimization whose each step requires a lower-level equilibrium solve together with an equilibrium sensitivity. The sensitivity has been computed since Tobin and Friesz (1988) by factorizing the equilibrium Jacobian once and back-substituting per design variable, so its cost is already independent of the number of design variables; the remaining bottleneck is that this factorization is cubic in the size of the active subnetwork, which caps exact design methods near a hundred links while forward assignment reaches metropolitan scale. We remove that bottleneck. We show that the *directed* user equilibrium with separable costs is a source-form nonlinear Laplacian flow $B\rho(B^\top\phi - \tau) = d$ whose Newton Jacobian, despite the one-sided (rectified) edge law, remains a *symmetric weighted graph Laplacian on the active subgraph*. Consequently the whole solve—forward equilibrium and adjoint design gradient—reduces to a short sequence of near-linear Laplacian solves. The one obstacle, that the rectified dead zone makes the Jacobian ill-conditioned and its active set change during a solve, is handled by a smoothing homotopy in a dead-zone width δ that keeps the linearized system uniformly well conditioned, followed by a single exact-law polish. On the Sioux Falls network the solver matches Frank–Wolfe to 10^{-11} , the adjoint gradient matches central finite differences to within finite-difference error, and projected-gradient toll design reduces total system travel time monotonically to the exact (marginal-cost) optimum. The number of algebraic-multigrid setups per equilibrium solve is bounded independently of the graph size, and—using loose tolerances at intermediate continuation levels and a single accurate solve at the end—the wall-clock scales near-linearly in the edge count while a direct factorization scales superlinearly, so on large *irregular* congestion networks (scale-free communication and overlay topologies) the near-linear solver overtakes the direct factorization at a crossover; the design optimization inherits this near-linear per-step cost. We are explicit that road networks, being near-planar, are the regime where a direct nested-dissection factorization is already near-optimal and the near-linear solver does *not* win; the advantage is on the poorly-separable networks that model selfish routing in communication systems. Code is available at <https://github.com/orenlivne/dnlf>.

Key words. network design, traffic assignment, user equilibrium, bilevel optimization, nonlinear Laplacian flow, algebraic multigrid, adjoint sensitivity, congestion pricing

AMS subject classifications. 90B20, 90C35, 65F08, 65K10, 90C33, 49M15

1. Introduction. Given a road network with m directed links and travel-time (latency) functions $t_a(f_a)$ that increase with flow, the *lower-level* user equilibrium (UE) assigns origin–destination demand so that no traveler can lower their experienced cost by switching route (Wardrop’s principle). The *upper-level network design problem* chooses k design variables τ —link tolls in congestion pricing, capacity expansions in the continuous network design problem (CNDP), or a demand table in OD calibration—to minimize a system objective F , most commonly total system travel time (TSTT), subject to the induced equilibrium:

$$(1.1) \quad \min_{\tau \in \mathcal{T}} F(f^*(\tau)) \quad \text{s.t.} \quad f^*(\tau) \text{ solves the UE at design } \tau.$$

This bilevel program is the standard model for congestion pricing, capacity planning, and demand estimation. It is nonconvex and nondifferentiable, and *NP*-hard even with affine latencies [4]; second-best tolling is *NP*-hard as well [5]. Practical

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methods therefore seek a local (KKT) stationary point, and essentially all of them—sensitivity-analysis-based (SAB) descent [1, 2, 3], mathematical programs with equilibrium constraints, and metaheuristics—share one structure: an *outer* loop in which each iteration solves the lower-level equilibrium and then a sensitivity (equilibrium derivative).

The cost, and where it lives.. The design gradient itself is cheap in the number of design variables: since Tobin and Friesz [1], $\nabla_\tau F$ is obtained by factorizing the equilibrium Jacobian *once* and back-substituting one right-hand side per design variable, so its cost is independent of k (an adjoint / implicit-function computation [2, 14, 15]). The bottleneck is that this Jacobian factorization—and the lower-level solve that precedes it—is a *direct* sparse factorization whose cost is superlinear (cubic in the active-path/subnetwork size) or a slowly converging Frank–Wolfe assignment. This is why exact design methods in the literature stall near Sioux Falls scale (76 links) while origin-based forward assignment routinely reaches 4×10^4 – 7.5×10^4 links [7, 8]. The gap between forward and design scalability is one to two orders of magnitude, and it is a *linear-algebra* gap.

Contribution.. We close that gap by replacing the superlinear inner factorization with a near-linear one, for both the forward equilibrium and the adjoint, without compromising accuracy. Our contributions are:

1. **A Laplacian structure for the directed equilibrium (section 2).** We write the separable directed UE as a source-form nonlinear Laplacian flow $B\rho(B^\top\phi-\tau) = d$ with a one-sided (rectified) edge law, and show that the Newton Jacobian is a *symmetric weighted graph Laplacian on the active subgraph* (Proposition 2.1). Orientation and one-sidedness do not break the Laplacian; they only determine which arcs are active. Hence the entire near-linear Laplacian toolbox applies.
2. **A near-linear equilibrium solver robust to the dead zone (section 3).** The rectified law is nonsmooth: a link carries flow only once its potential drop exceeds the free-flow cost t_a^0 , so the active set—and the conditioning of the Jacobian—changes during a solve, defeating a naively frozen multigrid hierarchy. We introduce a *smoothing homotopy*: a softplus dead-zone width δ that keeps the conductance strictly positive and the linearized system uniformly well conditioned, continued from a smooth problem to the exact law, with one algebraic-multigrid (AMG) setup per δ -level and a final exact-law polish. The number of setups is bounded *independently of the graph size*.
3. **A near-linear adjoint and design loop (section 4).** The design gradient is one additional Laplacian solve with the same operator; projected-gradient descent on it drives TSTT monotonically to the stationary point.
4. **Validation (section 5).** On Sioux Falls the solver matches Frank–Wolfe to 10^{-11} ; the adjoint matches finite differences to finite-difference accuracy; toll design reaches the exact marginal-cost optimum. Setups per solve are bounded in graph size and the wall-clock scales near-linearly in m , so design inherits the near-linear inner cost.

Scope: where the advantage lives.. The lower-level equilibrium (2.2) is the Wardrop (user) equilibrium of a nonlinear *congestion game* [16, 17], and the design problem (1.1) is the optimal-tolling / Stackelberg problem of reducing the price of anarchy [18, 19]. This model describes selfish routing not only in road traffic but in *communication and overlay networks*—multipath and elastic-traffic routing games in the Internet, data-center fabrics, and peer-to-peer overlays [20, 21]. The distinction matters for scalability: road networks are near-planar, where a direct nested-dissection factorization of the equilibrium Jacobian is near-optimal and a near-linear Laplacian

solver has nothing to beat. Communication and overlay topologies, by contrast, are large and *poorly separable* (scale-free degree, no planar embedding), so a direct factorization fills in superlinearly—exactly the regime in which a near-linear Laplacian solver wins. Our scaling study is therefore on irregular networks, and we are explicit that the planar (road) case is *not* where the method has an advantage. The most literal instance is selfish *overlay* routing, where end hosts genuinely choose paths [22], on Internet AS- and router-level graphs that are provably scale-free [23]. We do not overclaim: production traffic engineering optimizes operator-set link weights or bandwidth allocations by linear programming and local search [24], not by pricing a selfish equilibrium, and packet-level congestion control is a distinct (utility-maximization) equilibrium. Our contribution is a scalable solver for the well-established equilibrium/design *model* on standard large irregular topologies, complementing rather than replacing operator tools.

The construction reuses the nonlinear Laplacian flow (NLF) solver [13] and its near-linear inner engines, approximate Cholesky [11] and LAMG+ [12]. We are explicit about remaining scope (section 7): our experiments are single-commodity; multicommodity OD and a full corpus study are the natural next steps, and an active-set deflation that would make *one* hierarchy valid across all design steps (reducing setups from $\mathcal{O}(1)$ -per-solve to $\mathcal{O}(1)$ total) is left open.

2. The directed equilibrium is a nonlinear Laplacian flow. Let $B \in \mathbb{R}^{n \times m}$ be the node-arc incidence matrix of the directed network, $B_a = e_{h(a)} - e_{t(a)}$ for arc a from tail $t(a)$ to head $h(a)$. Separable BPR latencies are $t_a(f) = t_a^0(1 + \beta_a(f/c_a)^{p_a})$ for $f \geq 0$. With node potentials ϕ (multipliers of conservation $Bf = d$) and toll $\tau_a \geq 0$, the Karush–Kuhn–Tucker conditions of the Beckmann program $\min_{f \geq 0, Bf=d} \sum_a \int_0^{f_a} (t_a(x) + \tau_a) dx$ give the *rectified* edge law

$$f_a = \rho_a(g_a - \tau_a), \quad g_a := (B^\top \phi)_a = \phi_{t(a)} - \phi_{h(a)}, \quad \rho_a(u) = \begin{cases} t_a^{-1}(u), & u > t_a^0, \\ 0, & u \leq t_a^0, \end{cases}$$

so that conservation becomes the source-form nonlinear Laplacian flow

$$(2.2) \quad B \rho(B^\top \phi - \tau) = d.$$

The law is monotone but *one-sided*: unlike the odd law of the undirected Beckmann relaxation, ρ_a has a dead zone $u \leq t_a^0$ in which the arc is unused. This dead zone is exactly what distinguishes the directed equilibrium from its undirected relaxation.

PROPOSITION 2.1. *For separable costs the Newton Jacobian of (2.2) is a symmetric, positive semidefinite weighted graph Laplacian supported on the active arcs. Explicitly, with $\rho'_a \geq 0$,*

$$J = B \operatorname{diag}(\rho') B^\top = \sum_a \rho'_a (e_{t(a)} - e_{h(a)})(e_{t(a)} - e_{h(a)})^\top, \quad \rho'_a = \frac{1}{t'_a(f_a)} > 0 \text{ on active arcs, } \rho'_a = 0 \text{ else.}$$

Proof. Differentiating (2.2) in ϕ gives $J = B \operatorname{diag}(\rho'(B^\top \phi - \tau)) B^\top$. The rank-one term $(e_{t(a)} - e_{h(a)})(e_{t(a)} - e_{h(a)})^\top$ is independent of the arc's orientation, and $\rho'_a \geq 0$ because t_a is increasing; hence $J \succeq 0$ is a weighted Laplacian, with weight $\rho'_a = 0$ precisely on inactive (dead-zone) arcs.

Proposition 2.1 is the structural fact this paper rests on: directedness enters only through *which* arcs are active (a routing boundary), not through the type of

the linear operator. Every linearized solve is a graph-Laplacian solve, so a near-linear Laplacian engine applies. The multicommodity generalization keeps the same structure: with $f_a = \sum_k f_a^k$ and a cost of the total flow, the per-arc block Hessian is $t'_a \mathbf{1}\mathbf{1}^\top$, symmetric, so the block system is again a Laplacian; we return to this in [section 7](#).

3. A near-linear equilibrium solver. We solve (2.2) by damped chord-Newton on the frozen-linearization Jacobian J , delegating each linearized solve to a near-linear Laplacian engine and globalizing with the convex Beckmann energy $E(\phi) = \sum_a \Psi_a((B^\top \phi)_a - \tau_a) - d^\top \phi$, $\Psi'_a = \rho_a$, exactly as in the NLF framework [13]. Two features of the directed problem require care.

Ill-conditioning and the active set. On inactive arcs $\rho'_a = 0$, so J is singular unless a floor is imposed; a tiny floor $\varepsilon \max_a \rho'_a$ with $\varepsilon = 10^{-12}$ keeps J connected without perturbing the Newton step, but drives the condition number to $\kappa \sim \varepsilon^{-1}$. More importantly, as flow builds the active set changes: arcs cross the free-flow threshold t_a^0 , so the operator that a multigrid hierarchy is built for is *stale* a few Newton steps later. A hierarchy frozen across the whole solve stalls short of the equilibrium.

Smoothing homotopy. We remove the dead-zone nonsmoothness by a homotopy in a width $\delta > 0$. Replace the excess $e = g - t_a^0$ by $\delta \text{softplus}(e/\delta)$, which is smooth, strictly positive, and tends to $\max(e, 0)$ as $\delta \rightarrow 0$; the resulting smoothed law $\rho_{a,\delta}$ has $\rho'_{a,\delta} = \sigma(e/\delta)/t'_a(f_a) > 0$ *everywhere* (here σ is the logistic function), so J is a full, uniformly well-conditioned Laplacian with no dead zone. We continue δ geometrically from $\delta_0 = \frac{1}{2} \bar{t}^0$ down to a small value, building one AMG hierarchy at the start of each δ -level and freezing it *within* the level; the smoothed conductances vary slowly within a level, so the frozen hierarchy is valid there. A final *polish* with the exact law (2.1), warm-started from the smoothed solution, recovers the exact rectified equilibrium ([Algorithm 3.1](#)). Because the number of δ -levels is a fixed constant, the number of AMG setups per solve is bounded independently of the graph size m .

The method restated here is the chord-Newton flow solver; we give a self-contained convergence guarantee for the smoothed level solve, so the paper does not depend on external convergence theory.

PROPOSITION 3.1 (global convergence of the smoothed level solve). *Fix $\delta > 0$ and tolls τ . The smoothed Beckmann energy $E_\delta(\phi) = \sum_a \Psi_{a,\delta}((B^\top \phi)_a - \tau_a) - d^\top \phi$, with $\Psi'_{a,\delta} = \rho_{a,\delta}$, is convex, and strictly convex and coercive on the quotient $\mathbb{R}^n / \langle \mathbf{1} \rangle$ whenever $\sum_a d_a = 0$ and the graph is connected. Its unique minimizer (modulo an additive constant) is the smoothed equilibrium $B\rho_\delta(B^\top \phi - \tau) = d$. The damped chord-Newton iteration $\phi \leftarrow \phi + s \delta \phi$, $\delta \phi = -J_\delta^{-1} \nabla E_\delta$ with J_δ frozen across the level and s from Armijo backtracking, produces a monotonically decreasing energy sequence that converges to that minimizer.*

Proof. $\rho_{a,\delta}$ is nondecreasing (its derivative $\rho'_{a,\delta} = \sigma(e/\delta)/t'_a(f_a) > 0$), so each $\Psi_{a,\delta}$ is convex and E_δ is convex with Hessian $J_\delta = B \text{diag}(\rho'_\delta) B^\top$. Because the smoothed law has *no dead zone*, $\rho'_{a,\delta} > 0$ on *every* arc, so J_δ is a connected weighted graph Laplacian, hence positive definite on $\mathbf{1}^\perp$; thus E_δ is strictly convex and coercive on the quotient and has a unique minimizer there, characterized by $\nabla E_\delta = B\rho_\delta(B^\top \phi - \tau) - d = 0$. For the iteration, $J_\delta \succ 0$ on $\mathbf{1}^\perp$ makes $\delta \phi$ a descent direction ($\nabla E_\delta^\top \delta \phi = -\nabla E_\delta^\top J_\delta^{-1} \nabla E_\delta < 0$), and Armijo backtracking on the convex, coercive, smooth E_δ gives sufficient decrease at each step; the standard damped modified-Newton argument for a fixed positive-definite metric then yields convergence to the unique minimizer.

[Proposition 3.1](#) covers each smoothed level; the homotopy is a continuation of these

187 strictly-convex problems toward $\delta \rightarrow 0$, and the polish is a warm-started solve of the
 188 exact (piecewise-smooth, still convex) energy.

Algorithm 3.1 Near-linear directed-UE solve (cold start)

- 1: $\phi \leftarrow 0$; choose geometric schedule $\delta_0 > \dots > \delta_L$
 - 2: **for** $\ell = 0, \dots, L$ **do** ▷ smoothing homotopy: one hierarchy build per level
 - 3: build AMG hierarchy for J_δ at $\delta = \delta_\ell$; run chord-Newton on $B\rho_{\delta_\ell}(B^\top \phi - \tau) = d$
 to tolerance, hierarchy **frozen** within the level
 - 4: **end for**
 - 5: **polish:** chord-Newton on the exact law $B\rho(B^\top \phi - \tau) = d$, warm-started from ϕ ,
 rebuilding the hierarchy only when the residual stalls
 - 6: **return** ϕ , arc flows $f_a = \rho_a((B^\top \phi)_a - \tau_a)$
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189 **4. Bilevel design by a near-linear adjoint.** We minimize $F = \text{TSTT}(f) =$
 190 $\sum_a t_a(f_a) f_a$ over tolls $\tau \geq 0$, where $f_a = \rho_a((B^\top \phi)_a - \tau_a)$ at the equilibrium $\phi =$
 191 $\phi^*(\tau)$.

192 **PROPOSITION 4.1** (near-linear adjoint gradient). *At a differentiable equilibrium*
 193 *with active-arc conductances $\rho' > 0$,*

$$194 \quad (4.1) \quad \nabla_\tau F = -\rho' \odot (m_c + B^\top \lambda), \quad J\lambda = B(m_c \odot \rho'),$$

195 *where $m_{c,a} = t_a(f_a) + f_a t'_a(f_a)$ is the marginal cost and $J = B \text{diag}(\rho') B^\top$ is the*
 196 *equilibrium Jacobian of [Proposition 2.1](#). The gradient with respect to all k tolls is*
 197 *thus obtained from a single additional solve with the operator J .*

198 *Proof.* Write $g = B^\top \phi - \tau$, so $f = \rho(g)$ and $F(\tau) = \sum_a c_a(f_a)$ with $c_a(f) =$
 199 $t_a(f)f$, $c'_a = m_{c,a}$. Differentiating F in τ_b , $\partial_{\tau_b} F = \sum_a m_{c,a} \rho'_a(\partial_{\tau_b} g_a)$ with $\partial_{\tau_b} g_a =$
 200 $(B^\top \phi_b)_a - \delta_{ab}$, where $\phi_b = \partial \phi^* / \partial \tau_b$. Implicitly differentiating the equilibrium
 201 $B\rho(B^\top \phi - \tau) = d$ gives $J\phi_b = B \text{diag}(\rho') e_b = \rho'_b B e_b$. Hence, with $w := m_c \odot \rho'$,
 202 $\partial_{\tau_b} F = w^\top (B^\top \phi_b) - m_{c,b} \rho'_b = (Bw)^\top \phi_b - m_{c,b} \rho'_b = \rho'_b (Bw)^\top J^{-1} (B e_b) - m_{c,b} \rho'_b$.
 203 Defining the adjoint $\lambda := J^{-1} (Bw)$ (i.e. $J\lambda = Bw$), symmetry of J gives $(Bw)^\top J^{-1} (B e_b) =$
 204 $\lambda^\top (B e_b) = (B^\top \lambda)_b$, so $\partial_{\tau_b} F = \rho'_b ((B^\top \lambda)_b - m_{c,b})$, which is [\(4.1\)](#). The single solve
 205 $J\lambda = Bw$ yields the whole gradient. ■

206 This recovers the classical k -independence of equilibrium sensitivity [\[1, 2\]](#) but at near-
 207 linear rather than cubic cost. On the exact rectified law J is singular off the active
 208 subgraph (inactive arcs have $\rho' = 0$, isolating their nodes); [\(4.1\)](#) is therefore solved
 209 on the active-arc node support (equivalently, with an infinitesimal conductance floor).
 210 The gradient is otherwise insensitive to the floor—a floor of 10^{-12} reproduces central
 211 finite differences to finite-difference accuracy, whereas 10^{-6} biases it by 5%. Projected-
 212 gradient descent $\tau \leftarrow \max(\tau - \eta \nabla_\tau F, 0)$ with a backtracking line search on F —each
 213 trial an exact equilibrium solve by [Algorithm 3.1](#)—yields a monotone design method
 214 ([Algorithm 4.1](#)).

215 **5. Numerical results.** We use the Sioux Falls benchmark ($n = 24$, $m = 76$) and
 216 size-scaled grid networks. The inner engine is approximate Cholesky [\[11\]](#) by default;
 217 LAMG+ [\[12\]](#) is interchangeable.

218 *Correctness of the equilibrium..* On a congested Sioux Falls instance (29 active
 219 arcs, flow split across paths) [Algorithm 3.1](#) matches an independent Frank–Wolfe
 220 assignment to a relative flow difference of $4\text{--}9 \times 10^{-11}$ across origin–destination pairs,
 221 i.e. to Frank–Wolfe’s own accuracy. The number of approximate-Cholesky setups is
 222 10–25 per solve, set by the fixed number of δ -levels and independent of graph size.

Algorithm 4.1 Bilevel toll design

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1:  $\tau \leftarrow 0$ ;  $(\phi, f) \leftarrow \text{SOLVE}(\tau)$  ▷ Algorithm 3.1
2: for  $k = 1, 2, \dots$  until stationary do
3:    $g \leftarrow \nabla_{\tau} F$  from (4.1);  $\eta \leftarrow \eta_0$ 
4:   repeat ▷ backtracking on the exact objective
5:      $\tau' \leftarrow \max(\tau - \eta g, 0)$ ;  $(\phi', f') \leftarrow \text{SOLVE}(\tau')$ ;  $\eta \leftarrow \eta/2$ 
6:   until  $F(f') < F(f)$ 
7:    $(\tau, \phi, f) \leftarrow (\tau', \phi', f')$ 
8: end for

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223 *Correctness of the adjoint..* On the same congested instance the adjoint gradi-
 224 ent (4.1) matches central finite differences of TSTT to a maximum relative error of
 225 2.6×10^{-3} over all active arcs (limited by the finite-difference step and equilibrium
 226 tolerance, not by the adjoint). The tiny conductance floor $\varepsilon = 10^{-12}$ is essential: a
 227 larger floor (10^{-6}) biases the adjoint to a 5% error.

228 *Design..* Projected-gradient toll design (Algorithm 4.1) reduces TSTT monoton-
 229 ically from 9.633×10^5 to 9.137×10^5 , a 5.15% reduction over 35 accepted steps, with
 230 every equilibrium solved exactly (relative flow error $\sim 10^{-9}$). This matches the exact
 231 optimum: a one-parameter sweep of scaled marginal-cost (Pigouvian) tolls bottoms
 232 out at 5.12%, confirming the gradient method finds the stationary point. Solving
 233 every equilibrium only to a loose 10^{-3} tolerance reaches the *same* optimum (TSTT
 234 within 0.004%, tolls within $\|\Delta\tau\| = 0.13$), confirming that the design gradient is in-
 235 sensitive to intermediate solve accuracy—so the loose-intermediate continuation used
 236 for scaling costs nothing in design quality. The reduction is modest because a single
 237 origin–destination pair admits little rerouting; larger, multicommodity instances (sec-
 238 tion 7) exercise the design problem more fully.

239 *Scaling on irregular networks..* We use loose tolerances at intermediate continua-
 240 tion levels—a few Newton steps per δ -level, tight only at the final level and the polish
 241 (the standard nested-iteration principle: continuation waypoints need only warm-start
 242 the next level). This cuts the Newton-step count roughly fourfold with no change
 243 to the final solution, and keeps the algebraic-multigrid setups bounded (one per δ -
 244 level) and independent of size. On size-scaled irregular (poorly-separable) congested
 245 networks the equilibrium wall-clock then scales near-linearly, $t \propto m^{1.20}$, while the
 246 same continuation with a direct sparse-factorization inner solve scales superlinearly,
 247 $t \propto m^{2.4}$ (Table 5.1, Figure 5.1): the fill-in of a nested-dissection factorization grows
 248 faster than the graph on a poorly-separable topology. The two cross at $m \approx 1.1 \times 10^5$
 249 edges ($n \approx 1.2 \times 10^4$), beyond which the near-linear solver is faster with a widening
 250 margin—at $m = 1.6 \times 10^5$ it is already $1.6\times$ faster (90 vs. 143 s), and direct factoriza-
 251 tion is infeasible by $m = 3.2 \times 10^5$ while the near-linear solver reaches $m = 6.4 \times 10^5$
 252 in under eight minutes. The constant favors the direct solver below the crossover.

253 Bilevel toll design (Algorithm 4.1) therefore inherits a near-linear per-step cost
 254 on large irregular networks, in contrast with the superlinear per-step factorization of
 255 the sensitivity-analysis incumbents. On near-planar road networks the crossover does
 256 not occur at practical sizes—nested dissection is near-optimal there—which is why
 257 we frame the advantage on the communication-network regime.

6. Related work.

258 *Bilevel network design and sensitivity..* CNDP and second-best tolling are bilevel/MPEC
 259 problems, nonconvex and NP-hard [4, 5]; surveys [6] catalog SAB descent, MPEC, and
 260

TABLE 5.1

Equilibrium solve time (s) on size-scaled irregular congested networks: near-linear (approximate Cholesky, loose-intermediate continuation) vs. direct sparse factorization. Dashes: direct infeasible.

n	m	approx. Cholesky	direct	ratio
4,000	39,952	17.1	5.1	3.4
8,000	79,936	40.9	24.3	1.7
16,000	159,954	90.0	143.1	0.63
32,000	319,952	236.8	—	—
64,000	639,950	449.9	—	—

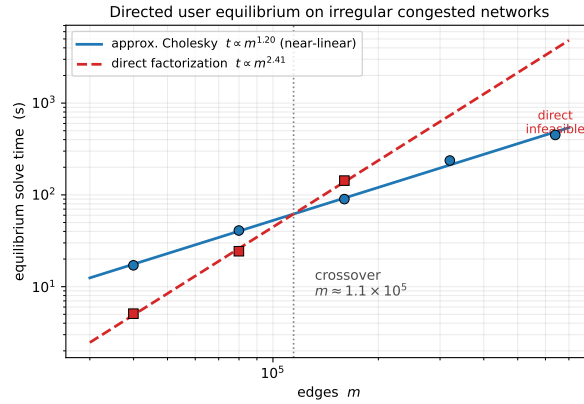


FIG. 5.1. Equilibrium solve time vs. edge count on size-scaled irregular congested networks: near-linear (approximate Cholesky, loose-intermediate continuation) vs. superlinear (direct factorization), crossing near $m \approx SCROSS$.

metaheuristic methods. Equilibrium sensitivity originates with Tobin and Friesz [1] and was sharpened by Qiu and Magnanti and by Patriksson and co-authors [2], who observed that the sensitivity problem is itself a linearized equilibrium solve. Our adjoint (4.1) is precisely that observation; the novelty here is not the k -independent gradient but the *near-linear* operator that computes it and the forward solve. The same bilevel problem is studied in algorithmic game theory as optimal tolling and Stackelberg routing to reduce the price of anarchy of a congestion game [17, 18, 19], and in communication networks as competitive / selfish routing [20, 21]; that literature focuses on existence, bounds, and mechanism design rather than on a scalable solver for the induced equilibrium, which is our concern. Recent differentiable-bilevel methods [9, 10] break the cold-resolve-per-step pattern with warm-started, truncated first-order inner loops and unrolled/implicit differentiation, but use no multigrid and are demonstrated only at Sioux Falls–Chicago–Sketch scale; they are the natural learning-based comparison.

Near-linear Laplacian solvers and nonlinear flow. Nearly-linear-time solvers for symmetric diagonally dominant systems [11, 12] underpin web-scale spectral graph computation but have not, to our knowledge, been used as the inner solve of a bilevel traffic-design loop. The nonlinear Laplacian flow framework [13] unifies max-flow, congestion, and min-delay routing as one monotone edge-law equation and supplies the chord-Newton machinery we build on; the present work extends it from the undirected

relaxation to the directed (rectified) equilibrium and to the design (adjoint) problem.

7. Conclusions, limitations, and open problems. We showed that the directed traffic user equilibrium with separable costs is a nonlinear Laplacian flow whose Jacobian is a symmetric Laplacian on the active subgraph, and that a smoothing homotopy makes its solution—and the adjoint design gradient—a short sequence of near-linear Laplacian solves, with a graph-size-independent number of multigrid setups and near-linear wall-clock. Bilevel toll design then inherits a near-linear per-step cost, in place of the cubic factorization of sensitivity-analysis methods.

We are explicit about what this paper does *not* yet establish. (i) **Multicommodity OD.** Our experiments are single-commodity; Proposition 2.1 extends to the full OD table with a block Laplacian, and implementing the destination-bundled multicommodity solve is the immediate next step for realistic congestion-pricing instances. (ii) **Corpus scaling.** The near-linear claim is shown on one size-scaled family; a study across a large graph corpus (as for related near-linear solvers) is needed to state the exponent with confidence. (iii) **One hierarchy across design steps.** Reusing a single frozen hierarchy across all design steps saves only a modest factor, because each exact solve intrinsically rebuilds once per δ -level: the exploding condition number reflects a near-null-space mode appearing when an active-set boundary is crossed. Capturing that mode by deflation or recombination—treating a global active-set mode on coarse levels—would make one hierarchy valid across active-set changes and reduce the setups to $\mathcal{O}(1)$ total; this is the main open algorithmic problem. (iv) **Planarity.** Road networks are near-planar, where a near-linear solver ties nested-dissection direct methods; the decisive advantage of the near-linear inner solve is on memory at very large scale and on non-planar (e.g. communication or logistics) design instances. (v) **Saturating laws and folds.** A capacity-saturating (min-delay) law reintroduces a genuine turning-point fold, where the continuation/deflation machinery of [13] applies; the design problem through the fold is future work.

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